Exploratory Design with Data

From Sensors to Stories

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Course Overview

01

March 25th

Sensing & Collecting

Build your RPi sensor. Choose a phenomenon as a team. Deploy and start collecting data in teams.

Today

02

April 1st

Analyzing & Storytelling

Clean your data in Excel. Use AI-assisted analysis. Find 2-3 story angles, each student will be responsible for their own final presentation

03

April 8th

Visualizing & Presenting

Refine your visualizations. Final individual presentations with open discussion.

What this course is

What it IS

A project-based arc: collect → analyze → visualize

An introduction to physical data collection and what it actually takes

Practice finding and telling a story with real, messy data

Built on tools from active DAAP research - you're working with the same hardware and methods used in published work

Graded on engagement and completion, not polish

What it ISN'T

A coding course - example scripts are provided

A hardware course - the electronics are set up; focus on data

Expected to be perfect - messy data with honest documentation is the standard

About learning one specific visualization tool – try a few tools and use what works best for your story

Where this comes from

This course is built on active dissertation research at DAAP. The tools you'll use are the same open-source hardware and analysis methods used in published work on pedestrian experience and environmental sensing in urban spaces.

Raspberry Pi toolkit

Presented at IRIS Conference Fall 2025. Published on GitHub and Zenodo. Three device configurations: fixed sensor, mobile, and head-mounted.

AI Assisted Analysis Tool

Local LLM-based analysis via Ollama. No data sent externally. Developed to process large text and image datasets in planning research.

Workshop with planning students

Prior workshop in Dr. Cho's Smart Cities course gave students hands-on sensor experience. This masterclass is the second data point.

Resources & Code

Sensor Workshop

github.com/henrylevesque/sensor_workshop

All workshop scripts: test files, data loggers, camera capture, wiring guide, quick-start instructions.

RPi Data Collection

github.com/henrylevesque/rpi_data_collection

More configurations including mobile and head-mounted setups. Reference for additional sensor types and multi-sensor scripts.

AI Assisted Analysis

github.com/henrylevesque/AI_Assisted_Analysis_Tool

Local LLM text and image analysis. Used in Session 2 for pattern extraction. Runs entirely on your machine.

Visualization Tools

Flourish (flourish.studio) · Rawgraphs (rawgraphs.io) · p5.js (p5js.org) · D3.js

Zenodo DOI: 10.5281/zenodo.14932653 (AI Assisted Analysis Tool)

Today - Session 1

9:30 **Intro + context**

Why physical data collection? What makes it different from secondary data? Overview of the full arc.

9:55 **Demo**

RPi setup, sensor wiring, first data run. SSH, running scripts, exporting your first Excel file.

10:20 **Team formation + phenomenon selection**

Groups of 2. Pick something you're curious about measuring or visualizing. Choose your sensor(s).

10:40 **Short break**

10:50 **Activity**

Get your sensor running. Confirm data is logging. Connect to a 2.4GHz hotspot before leaving - you'll need remote access between sessions.

12:00 **Wrap + deploy**

What you're taking home. Expectations for Session 2. Questions.

The Hardware

What's in your kit

- Raspberry Pi Zero 2W - a single-board computer about the size of a stick of gum
- PiSugar battery - powers the Pi without a wall outlet (a few hours; use a power bank or outlet for extended deployment)
- Sensors - temperature/humidity (DHT11), PIR motion, analog sound, camera module, and others
- Jumper wires, SD card (pre-flashed with scripts)
- 3D printed enclosure - designed for this workshop, prints available on Printables.com

Things worth knowing now

- RPi Zero only connects to 2.4GHz WiFi - not the standard 5GHz network
- Before you leave today: connect it to your phone hotspot or 2.4GHz WiFi so you can reach it remotely
- Check your wiring before powering on. If something gets hot or smells wrong, unplug it
- Shut the Pi down through the OS before cutting power - just pulling the plug can corrupt the SD card
- You don't need to understand every line of code. The scripts are there to run

Available Sensors

Confirmed working - scripts and wiring guides in the repo:

Temperature + Humidity

DHT11 · GPIO 4
Outputs: temp (°C), humidity (%)
Best for: indoor climate, outdoor environment, space comparison

PIR Motion

HC-SR501 · GPIO 23
Outputs: motion detected (True/False)
Best for: activity in a space, occupancy patterns, event detection

Analog Sound

MAX4466 mic + ADS1115 ADC · I2C
Outputs: sound level in dB
Best for: ambient noise, activity events, before/after comparison

Also available (ask if interested, full list in the GitHub Repo):

Camera (Pi Camera module) · Photocell (light sensor) · GPS (low signal strength)

Connecting to your Pi - Quick Reference

Connect (SSH)

```
# Mac / Linux
ssh pi@rpizero2w1.local

# Windows (PowerShell)
ssh pi@rpizero2w1

# Default password
raspberry
```

Download your data (SCP)

```
# Mac / Linux
scp pi@rpizero2w1.local:/home/pi/
temperature_data.xlsx ~/Desktop/

# Windows
scp pi@rpizero2w1:/home/pi/
temperature_data.xlsx C:\Desktop\
```

Run a script

```
python3 temperature_test.py
# Press Ctrl+C to stop
```

WiFi note

The Pi Zero only connects to 2.4GHz networks. Institutional WiFi often blocks it. Use your phone hotspot or a personal 2.4GHz router. You must configure WiFi before leaving class today - you won't be able to reach the Pi remotely otherwise.

Between now and April 1

1

Deploy your sensor

3-5 days of collection depending on your phenomenon. (can collect for short periods over multiple days or locations) Make sure the Pi is connected to WiFi you can reach.

2

Monitor occasionally

SSH in to check the script is still running and the data file is growing. If something stops, restart the script.

3

Retrieve your data

Download the Excel or CSV file using SCP before Session 2 on April 1st.

4

Document what happened

If something didn't work, write down what you tried and what you observed. That is legitimate content. Hardware is unpredictable and example datasets will be provided.

Assessment

Assessment is based on engagement and completion, not polish.

Sensor & data collection

You attempted to collect real data with a physical sensor. Messy data with good documentation is fine.

Data analysis & story

You identified something in your data worth communicating and can explain what it shows - and what it doesn't.

Visualization

At least three visualizations that support the story and are legible to someone who wasn't there.

Presentation

Presented clearly. Reflected on what you'd do differently. Open discussion at the end of Session 3..

AI in this course

AI tools are part of the course and will be demonstrated, but are not a requirement

Code generation

The sensor scripts were developed with AI assistance. Showing you how to use AI to write and fix code is explicit pedagogy - being a critical consumer of AI-generated code is a more useful skill than writing it from scratch.

Data analysis (Session 2)

The AI Assisted Analysis Tool will extract patterns from images or text. All analysis runs locally - no data is sent to external servers. When AI shapes your analysis, document it: which tool, what prompt, what it found vs. what you found manually. The AI Assisted Analysis Tool can append metadata to the bottom of your output file.

Visualization ideation

AI can generate visualization concepts and starter code (p5.js sketches in JS). Some of those will have errors. Evaluating and fixing them is part of the exercise.

Questions?

Let's get into it.